	Low	Moderate	High	Extreme
HA	+85.10	+62.84	+20.25	+11.98
SVR	+51.02	+31.24	+11.93	+7.98
VAR	+59.24	+39.83	+17.05	+8.11
DCRNN	+60.81	+17.46	+4.89	+4.98
STGCN	-1.06	+0.95	+1.22	+2.74
GWNet	-0.99	+2.44	+2.16	+3.72
STNorm	-4.53	-1.24	-1.21	+0.36
DSTAGNN	+81.88	+58.45	+20.53	+5.25
PDFormer	+78.48	+64.15	+52.07	+58.45
STWave	-4.86	-2.01	-1.50	+0.36
PatchTST	+70.02	+12.48	+7.45	+9.80
TimeMixer	+71.07	+17.32	+10.50	+9.85
STDN	+85.49	+70.43	+32.57	+5.55
MetaDG	+84.72	+66.55	+33.15	+12.43
HyperD	-0.70	+0.66	+0.24	+3.29
RAST	-0.65	+1.78	+2.45	+4.57
CMSRN MAE	**0.161**	**0.344**	**0.705**	**1.340**
Valid n	59,917	129,101	182,079	128,448

(a) Chengdu · inflow

	Low	Moderate	High	Extreme
HA	+86.85	+58.46	+15.35	+9.02
SVR	+53.48	+25.66	+8.83	+5.36
VAR	+62.29	+33.20	+12.24	+5.98
DCRNN	+59.90	+12.07	+2.32	+2.10
STGCN	-1.24	-0.31	+0.25	+1.71
GWNet	-2.31	+0.37	+0.53	+1.88
STNorm	-2.75	-1.77	-1.48	-0.18
DSTAGNN	+84.77	+55.60	+18.54	+4.60
PDFormer	+80.34	+61.19	+50.46	+54.71
STWave	-3.08	-2.17	-1.23	-0.60
PatchTST	+72.28	+9.07	+5.09	+7.79
TimeMixer	+73.16	+14.00	+8.28	+8.39
STDN	+87.35	+66.29	+25.88	+4.26
MetaDG	+86.66	+63.20	+28.82	+10.24
HyperD	-1.12	-0.59	-0.33	+1.11
RAST	-0.87	+1.05	+1.38	+2.71
CMSRN MAE	**0.139**	**0.380**	**0.765**	**1.246**
Valid n	58,257	150,159	182,619	135,869

(b) Chengdu · outflow

	Low	Moderate	High	Extreme
HA	+82.70	+39.24	+12.29	+13.26
SVR	+50.12	+17.76	+8.41	+7.68
VAR	+60.50	+25.32	+10.84	+7.83
DCRNN	+10.07	+2.22	+1.90	+3.57
STGCN	+2.16	+1.17	+1.36	+4.04
GWNet	+1.16	+1.14	+1.56	+3.58
STNorm	-0.09	-1.10	-1.20	+0.45
DSTAGNN	+78.67	+32.96	+8.89	+5.15
PDFormer	+74.49	+52.53	+48.80	+57.73
STWave	-0.94	-1.01	-1.14	+0.24
PatchTST	+57.94	+6.41	+5.36	+10.20
TimeMixer	+58.88	+10.61	+8.17	+10.25
STDN	+84.56	+53.94	+25.57	+15.35
MetaDG	+83.14	+49.77	+24.81	+15.15
HyperD	+5.02	+0.78	+0.25	+2.83
RAST	+0.99	+0.90	+1.43	+3.14
CMSRN MAE	**0.177**	**0.536**	**0.851**	**1.296**
Valid n	92,199	223,697	222,486	115,210

(c) Shanghai · inflow

	Low	Moderate	High	Extreme
HA	+84.45	+37.87	+12.08	+12.06
SVR	+53.58	+16.45	+8.12	+7.17
VAR	+64.91	+23.61	+10.76	+7.36
DCRNN	+11.08	+2.06	+1.62	+3.42
STGCN	+2.06	+0.72	+1.10	+2.99
GWNet	+0.78	+0.60	+1.14	+3.56
STNorm	+0.07	-1.17	-1.21	+0.37
DSTAGNN	+81.23	+32.08	+9.31	+3.61
PDFormer	+76.69	+52.17	+49.93	+58.76
STWave	-0.35	-1.15	-1.03	+0.24
PatchTST	+61.53	+5.99	+5.12	+9.52
TimeMixer	+62.23	+9.82	+8.06	+9.74
STDN	+85.69	+43.02	+13.98	+6.11
MetaDG	+84.75	+49.60	+28.02	+16.62
HyperD	+1.94	+0.42	+0.04	+1.14
RAST	+2.31	+1.15	+1.55	+4.06
CMSRN MAE	**0.158**	**0.553**	**0.870**	**1.480**
Valid n	84,381	216,920	227,734	110,618

(d) Shanghai · outflow

